

INTERNAL ENGINEERING DOCUMENT

Model-Wise Software Troubleshooting Guide

Production Jigs — SmartManufacturingSuite

DOCUMENT VERSION

2.0

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MODELS COVERED

FG23 · 3PH LTCT · 3PH WC · IMG STG1 · IMG STG2

REPOSITORY

lalit-tak/Production_jigs

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Model-Wise Software Troubleshooting Guideline

Production Jigs — SmartManufacturingSuite

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Table of Contents

1. [Document Overview](#)
2. [Repository Analysis](#)
3. [Model Identification Matrix](#)
4. [Model-Wise Troubleshooting Guide](#)
5. 4.1 [FG23 — NIC RF Test Jig](#)
6. 4.2 [3PH LTCT — 3-Phase LTCT Test Jig](#)
7. 4.3 [3PH WC — 3-Phase Whole-Current Test Jig](#)
8. 4.4 [IMG Stage 1 — Firmware Flash & Provisioning Jig](#)
9. 4.5 [IMG Stage 2 — RF & Power Test Jig \(8CH / 16CH\)](#)
10. [Software Logs Analysis Guide](#)
11. [Diagnostic Flowcharts](#)
12. [Root Cause Analysis \(RCA\) Matrix](#)
13. [Recovery Procedures](#)
14. [Preventive Maintenance Checklist](#)
15. [Version Compatibility Matrix](#)
16. [Deployment & Upgrade Troubleshooting](#)
17. [Emergency Troubleshooting Quick Reference](#)
18. [Recommendations](#)

1. Document Overview

1.1 Purpose

This document provides a comprehensive, model-specific software troubleshooting reference for production engineers, NPI engineers, software developers, test engineers, maintenance teams, and shop-floor operators working with the Production Jigs platform ([SmartManufacturingSuite](#)). All information is derived directly from the repository source code, branch configurations, and actual test scripts.

1.2 Scope

This guide covers all jig models currently present in the [Production_jigs](#) repository:

- FG23 NIC RF Test Jig (Branches: [FG_23_DEC_CH_8](#) , [FG_23_DEC_CH_16](#))
- 3PH LTCT Test Jig (Branches: [3PH_LTCT_TOKYO](#) , [3PH_LTCT_ZETTOWN](#))
- 3PH WC Test Jig (Branches: [3PH_WC_TOKYO](#) , [3PH_WC_zettown](#))
- IMG Stage 1 Jig (Branches: [IMG_STAGE_1_DEC_1_CH_16](#) , [IMG_STAGE_1_DEC_2_CH_8](#))
- IMG Stage 2 Jig (Branches: [IMG_STAGE_2_1_DEC_8_CH](#) , [IMG_STAGE_2_1_DEC_16_CH](#))

1.3 Supported Models

Model Code	Full Name	Jig Purpose
FG23	Smart Meter NIC RF Test	RF radio link test for NIC modules
3PH-LTCT	3-Phase LTCT Meter Test	Live test of 3-phase voltage, current, metering
3PH-WC	3-Phase Whole-Current Meter Test	Whole-current meter functional test
IMG-STG1	IMG Module Stage 1	Firmware flash + optical + provisioning
IMG-STG2	IMG Module Stage 2	RF RSSI, supercapacitor, power & EMS test

1.5 Software Architecture Overview

```

SmartManufacturingSuite.py (Main GUI – Tkinter)
├── FirmwareFlash.py           ← Gang firmware programmer (IMG Stage 1, LTCT, WC)
│   └── GangProARM-FPAsel.dll  ← ctypes binding for Segger/gang programmer hardware
├── scripts/utils/communication.py ← Serial packet protocol (0xAA frame, 9600 baud)
├── scripts/tests/             ← Per-test Python modules (subprocess-launched)
│   ├── live_version_test.py   ← Firmware version verification (LTCT, WC)
│   ├── rtc_calibration_test.py ← RTC drift calibration (LTCT, WC)
│   ├── digital_input/
│   │   ├── voltage_test.py    ← 3-phase R/Y/B voltage decode
│   │   ├── current_test.py    ← 3-phase R/Y/B/N current decode
│   │   ├── magnet_status_test.py ← Magnetic tamper sensor
│   │   ├── cover_open_test.py ← Cover tamper switch
│   │   ├── push_button1/2_test.py ← Push button debounce
│   │   ├── memory_test.py     ← EEPROM/flash memory test
│   │   ├── nic_status_test.py ← NIC module presence
│   │   ├── rx_tx_status_test.py ← UART loopback
│   │   ├── md_reset_button_test.py ← Manual demand reset
│   │   ├── get_efuse_mode.py  ← eFuse security read
│   │   └── set_efuse_mode.py  ← eFuse security write
│   ├── IMG_Provisioning_Testing_Scripts/
│   │   ├── provision_img_parameters.py ← Write manufacturing params (Wirepas mesh)
│   │   ├── read_img_parameters.py     ← Verify provisioned params
│   │   ├── wirepas_provisioning.py    ← Wirepas network provisioning
│   │   ├── testing.py                ← RF RSSI test orchestrator
│   │   └── factory_mode.py           ← Factory mode activation
│   ├── staticDO.py                 ← Advantech BDAQ Digital Output (IMG Stage 1/2)
│   └── InstantAI.py                ← Advantech BDAQ Analog Input (IMG Stage 1/2)
├── scripts/Automation/BDAQ/        ← Full Advantech BDAQ Python SDK (DO/DI/AI/AO)
├── firmware_version_integration.py ← DLMS/COSEM serial firmware version check (FG23)
├── errors_dll.py                   ← GangProARM DLL error code lookup table
├── serial_config.enc               ← Encrypted COM port configuration (Fernet/PBKDF2)
├── config.txt                      ← Plain-text COM port config (LTCT/WC jigs)
└── firmwareflash_config.json       ← Firmware filenames config (WC branch)

```

1.6 System Dependencies

Dependency	Version/Notes	Used By
Python	3.12 / 3.13	All branches
pyserial	Latest	All serial communication
openpyxl	Latest	Test result Excel files
Pillow (PIL)	Latest	GUI logo/branding rendering
tkinter	stdlib	GUI framework

Dependency	Version/Notes	Used By
cryptography (Fernet)	Latest	serial_config.enc decryption
psutil	Latest	Process kill / timeout management
crcmod	Latest	Packet CRC validation
pythoncom / win32gui / win32con / win32com	pywin32	IMG Stage 2 (Windows COM)
Advantech BDAQ	USB-4704 / USB-4716	IMG Stage 1/2 DAQ hardware
GangProARM-FPAsel.dll	Segger OEM	Firmware flash (LTCT, WC, IMG Stage 1)

2. Repository Analysis

2.1 Branch-to-Model Mapping

```

Production_jigs (GitHub)
├── master                ← IMG Stage 2 (V1.0.0.3 binary + serial_config.enc only)
├── 3PH_LTCT_TOKYO       ← 3PH LTCT – Tokyo factory, full suite
├── 3PH_LTCT_ZETTOWN     ← 3PH LTCT – Zettown factory variant
├── 3PH_WC_TOKYO        ← 3PH WC – Tokyo factory, firmwareflash_config.json
├── 3PH_WC_zettown      ← 3PH WC – Zettown factory variant
├── FG_23_DEC_CH_8      ← FG23 NIC RF – 8-channel jig
├── FG_23_DEC_CH_16     ← FG23 NIC RF – 16-channel jig
├── IMG_STAGE_1_DEC_1_CH_16 ← IMG Stage 1 – 16CH DAQ layout
├── IMG_STAGE_1_DEC_2_CH_8  ← IMG Stage 1 – 8CH DAQ layout (alt)
├── IMG_STAGE_2_1_DEC_8_CH  ← IMG Stage 2 – 8CH DAQ layout
└── IMG_STAGE_2_1_DEC_16_CH ← IMG Stage 2 – 16CH DAQ layout

```

2.2 Key Software Modules

Module	File	Branch(es)	Purpose
Main GUI	<code>SmartManufacturingSuite.py</code>	All	Tkinter UI, orchestration, station management
Legacy Main	<code>SMS.py</code>	FG23, IMG-STG2	Earlier GUI variant
Firmware Flash	<code>FirmwareFlash.py</code>	LTCT, WC, IMG-STG1	GangProARM DLL wrapper
Main Firmware Flash	<code>MainFirmwareFlash.py</code>	FG23, IMG-STG2	Alternate flash controller
Communication	<code>scripts/utils/communication.py</code>	LTCT, WC	0xAA-framed serial protocol
Firmware Version	<code>firmware_version_integration.py</code>	FG23, IMG-STG2	DLMS/COSEM 0x7E-framed check
Error Codes	<code>errors_dll.py</code>	FG23, IMG-STG2	GangProARM status code map

Module	File	Branch(es)	Purpose
IMG Provision	<code>provision_img_parameters.py</code>	IMG-STG2	Wirepas parameter write
BDAQ SDK	<code>scripts/Automation/BDAQ/</code>	IMG-STG1/2	Advantech DAQ Python API
DI/DO Tests	<code>scripts/tests/staticDO.py</code>	IMG-STG1/2	USB-4716 digital output
AI Read	<code>scripts/tests/InstantAI.py</code>	IMG-STG1/2	USB-4704 analog input

2.3 Configuration Files

File	Location	Format	Description
<code>serial_config.enc</code>	Root	Fernet-encrypted	COM port assignments (password: <code>Polaris@1234</code>)
<code>config.txt</code>	Root	KEY=VALUE	COM port config (LTCT/WC plain-text)
<code>firmwareflash_config.json</code>	Root	JSON	Firmware filenames for flash
<code>FPA-setup.ini</code>	<code>firmware_flash_files/</code>	INI	GangPro adapter initialization
<code>*GangConfig.cfg</code>	<code>ConfigFiles/</code>	Binary/text	Gang programmer channel config
<code>RF_criteria.json</code>	Root (FG23/IMG-STG2)	JSON	RSSI pass/fail thresholds
<code>Load_channel_data.json</code>	Root (FG23)	JSON	Channel data mapping
<code>requirements.txt</code>	Root	pip format	Python dependencies

2.4 Hardware Interfaces

Interface	Hardware	Protocol	Branches
Serial COM	Meter DUT	Custom 0xAA frame, 9600 baud	LTCT, WC
Serial COM	NIC scanner / barcode	Serial trigger <code>*T</code> , 9600 baud	FG23, IMG
Serial COM	Meter firmware	DLMS/COSEM 0x7E frame, 19200 baud	FG23
USB-CDC	FPA Adapter	GangProARM DLL (ctypes)	LTCT, WC, IMG-STG1
USB-DAQ DO	Advantech USB-4716	BDAQ SDK	IMG-STG1/2
USB-DAQ AI	Advantech USB-4704	BDAQ SDK	IMG-STG1/2
Wirepas RF	NIC Module	UART provisioning	IMG-STG2
Optical	Meter optical port	Serial, 9600 baud	IMG-STG1

2.5 Log & Output Files

Output Type	Location	Format	Notes
		Text	

Output Type	Location	Format	Notes
Program log	Log_files/Log_file_YYYY-MM-DD/YYYY-MM-DD_program_log.txt		DEBUG level; daily rotation
Action log	Log_files/Log_file_YYYY-MM-DD/action_log.txt	Text	Operator actions with serial number
Test result	Test Result Excel File/Test_Result_YYYY-MM-DD/YYYY-MM-DD_*.xlsx	Excel	Per-station result rows
EMS result	EMS Test Result/YYYY-MM-DD/<SERIAL>_PASS.txt	Text	EMS-format per-device result
Detailed log	scripts/tests/IMG_Provisioning_Testing_Scripts/Detailed Test Result/	Text	IMG provisioning detail

3. Model Identification Matrix

Model Name	Software Module	Jig Type	Production Stage	Key Dependencies	Branch
FG23 (8CH)	SmartManufacturingSuite.py + SMS.py	NIC RF Test	Module sub-assembly	pyserial, openpyxl, GangProARM DLL, firmware_version_integration	FG_23_DEC_CH_8
FG23 (16CH)	SmartManufacturingSuite.py + SMS.py	NIC RF Test	Module sub-assembly	pyserial, openpyxl, GangProARM DLL, firmware_version_integration	FG_23_DEC_CH_16
3PH LTCT Tokyo	SmartManufacturingSuite.py	3-Phase Meter Test	Final test	pyserial, openpyxl, GangProARM DLL, communication.py, digital_input tests	3PH_LTCT_TOKYO
3PH LTCT Zettown	SmartManufacturingSuite.py	3-Phase Meter Test	Final test	pyserial, openpyxl, config.txt	3PH_LTCT_ZETTOWN
3PH WC Tokyo	SmartManufacturingSuite.py	Whole-Current Meter Test	Final test	pyserial, openpyxl, GangProARM DLL, firmwareflash_config.json	3PH_WC_TOKYO
3PH WC Zettown	SmartManufacturingSuite.py	Whole-Current Meter Test	Final test	pyserial, openpyxl	3PH_WC_zettown
IMG Stage 1 (16CH)	SmartManufacturingSuite.py	Firmware Flash + Provision	Module assembly	BDAQ, GangProARM DLL, staticDO, InstantAI, serial_config.enc	IMG_STAGE_1_DEC_1_0
IMG Stage 1 (8CH)	SmartManufacturingSuite.py	Firmware Flash + Provision	Module assembly	BDAQ, GangProARM DLL, staticDO, InstantAI, serial_config.enc	IMG_STAGE_1_DEC_2_0

Model Name	Software Module	Jig Type	Production Stage	Key Dependencies	Branch
IMG Stage 2 (8CH)	SmartManufacturingSuite.py	RF + Power Test	Module final test	BDAQ, pywin32, staticDO, InstantAI, provision scripts, serial_config.enc	IMG_STAGE_2_1_DEC_8
IMG Stage 2 (16CH)	SmartManufacturingSuite.py	RF + Power Test	Module final test	BDAQ, pywin32, staticDO, InstantAI, provision scripts, serial_config.enc	IMG_STAGE_2_1_DEC_16

4. Model-Wise Troubleshooting Guide

4.1 FG23 — NIC RF Test Jig

Purpose

The FG23 jig validates NIC (Network Interface Card) radio module sub-assemblies. It performs firmware flashing (test + main firmware), RF link quality testing (RX/TX RSSI), external flash test, firmware version verification via DLMS/COSEM protocol (0x7E frames at 19200 baud), and supercap voltage measurement. Supports **6 stations** simultaneously. Two variants: 8-channel DAQ (`FG_23_DEC_CH_8`) and 16-channel DAQ (`FG_23_DEC_CH_16`).

Test Sequence (All FG23 Variants — 11 Steps)

Step	Test Name	Action
1	Mapped with COM Port	COM port assigned to station
2	Module Supply On	DO write — powers module
3	LDO Voltage	AI read — LDO rail measurement
4	Super Cap Charge Voltage	AI read — supercap charge measurement
5	Test Firmware Download	Firmware flash via GangProARM
6	External Flash Test	Verify external SPI flash
7	RSSI Test	RF RX/TX RSSI measurement (0x7E, 19200 baud)
8	Main Firmware Download	Main firmware flash via GangProARM
9	Main Firmware Verification	DLMS version check — <code>packet[4] == 0x15</code>
10	Module Supply Off	DO write — powers down module
11	Super Cap Discharge Voltage	AI read — supercap discharge measurement

DAQ Digital Output Map — FG23 8CH (`FG_23_DEC_CH_8`)

`staticDO.py` invoked with only `--inputVal` ; device description hardcoded inside `staticDO.py` .

DO Hex Value	Test Step	Meaning
<code>0x01</code>	Module Supply On	Turn module supply ON
<code>0x02</code>	Module Reset (between firmware flashes)	Temporary reset / supply off
<code>0x05</code>	Module Supply Off	Turn module supply OFF

DAQ Digital Output Map — FG23 16CH (`FG_23_DEC_CH_16`)

Device: `USB-4716,BID#0` (from `serial_config.enc` key `DAC_COM`). DO values are decimal strings (`'06'` , `'05'`).

DO Hex Value	Test Step	Meaning
<code>0x06</code>	Module Supply On	Turn ON module supply
<code>0x05</code>	Module Supply Off / Firmware reset cycles	Turn OFF module supply (also used between firmware flash steps)

Critical: FG23 8CH and 16CH use **different DO values** for supply on (`0x01` vs `0x06`). Deploying the wrong branch on the wrong card will apply incorrect power states.

DAQ AI Channel Map — FG23 8CH (`FG_23_DEC_CH_8`)

Card: `USB-4716,BID#` — channel formulas: LDO = `station_num + 4` ; SuperCap = `station_num - 1`

Station	LDO Voltage	SuperCap Charge	SuperCap Discharge
1	CH5	CH0	CH0
2	CH6	CH1	CH1
3	CH7	CH2	CH2
4	CH8	CH3	CH3
5	CH9	CH4	CH4
6	CH10	CH5	CH5

DAQ AI Channel Map — FG23 16CH (`FG_23_DEC_CH_16`)

Card: `USB-4716,BID#0` — channel formulas: LDO = `station_num + 6` ; SuperCap = `station_num - 1` . Requires 16-channel AI card (LDO channels reach CH12).

Station	LDO Voltage	SuperCap Charge	SuperCap Discharge
1	CH7	CH0	CH0
2	CH8	CH1	CH1
3	CH9	CH2	CH2
4	CH10	CH3	CH3
5	CH11	CH4	CH4
6	CH12	CH5	CH5

Excel Column Schema (`YYYY-MM-DD_NIC_RF_TEST_RESULT.xlsx`) — 23 Columns

#	Column Header	#	Column Header
1	Station Number	13	RF Status
2	NIC serial number	14	External Flash
3	Start Time	15	Main Firmware Test
4	End Time	16	Firmware Version
5	Module Supply On	17	Expected Firmware Version
6	LDO Voltage	18	Firmware Version Test Status
7	LDO Voltage Status	19	Module Supply Off
8	Supercap Charging Voltage	20	Supercap Discharging Voltage
9	Supercap Charging Voltage Status	21	Supercap Discharging Voltage Status
10	Test Firmware Test	22	Overall Result
11	RX RSSI	23	Total Time
12	TX RSSI		

Result files: `Test Result Excel File/Test_Result_YYYY-MM-DD/YYYY-MM-DD_NIC_RF_TEST_RESULT.xlsx`
 EMS files: `EMS Test Result/YYYY-MM-DD/{serial_number}_{PASS|FAIL}.txt`

Firmware Configuration

Branch	Config Source	Test Firmware Key	Main Firmware Key	Gang Config
8CH	<code>config.txt</code> (1=COM3, 2=COM4, ...)		Loaded per <code>config.txt</code>	<code>ConfigFiles/GangConfig.cfg</code>

Branch	Config Source	Test Firmware Key	Main Firmware Key	Gang Config
		Loaded per <code>config.txt</code> station map		
16CH	<code>serial_config.enc</code>	<code>Test_Firmware</code>	<code>Main_Firmware</code>	<code>Gang_Programming_Code_File</code>

Default expected firmware version: `6.0.0.0` (from `ConfigFiles/firmware_config.txt`). All stations will fail Main Firmware Verification if the actual flashed version differs.

Startup Flow

1. Python runtime launches `SmartManufacturingSuite.py`
2. `os.makedirs()` creates `Test Result Excel File/`, `Log files/` directories
3. Logging initializes to `Log files/Log_file_YYYY-MM-DD/YYYY-MM-DD_program_log.txt`
4. Tkinter GUI window opens with station grid (6 stations)
5. 16CH: `serial_config.enc` decrypted (password `Polaris@1234`) → COM ports, firmware filenames, DO device loaded
6. 8CH: `config.txt` parsed for `1=COMx` station-to-COM mappings
7. BDaq AI/DO cards initialized
8. Scanner COM ports initialized — trigger: `*T` command at 9600 baud
9. Operator scans barcode → serial number captured → test sequence dispatched

Common Errors

Error Message	Possible Cause	Impact	Resolution
<code>Failed to open COM port: [Errno 13] Permission denied</code>	Another process holds port	Station blocked	Close competing application; restart jig software
<code>Failed to open COM port: [Errno 2] No such file or directory</code>	COM port changed after USB re-plug	Station offline	Check Device Manager; update COM port in config
<code>No response received (timeout).</code>	DUT not powered; DLMS frame mismatch	Version check fails	Verify DUT power; confirm firmware version ≥ expected
<code>Invalid packet received (10th byte is not 0x15). Ignoring...</code>	Wrong DUT; corrupted DLMS response	Version check fails	Verify correct meter connected; check cable/connectors
<code>STATUS_JTAG_COMM_ERROR (523)</code>	FPA adapter not detected	Firmware flash blocked	Check USB cable to programmer; verify DLL at <code>firmware_flash_files/GangProARM-FPAse1.dll</code>
<code>STATUS_FW_VERIFICATION_ERROR (526)</code>	Flash verify mismatch	Firmware write failed	Re-flash; check firmware binary integrity

Communication Failures

Scenario	Symptoms	Root Cause	Fix
Scanner port silent	No serial number on screen after scan	*T trigger not ACKed; baud mismatch	Verify scanner at 9600 baud, no parity, 1 stop bit
DLMS timeout	No response received (timeout) in log	DUT firmware not running; wrong baud	Confirm SERIAL_BAUDRATE = 19200 ; confirm DUT powered
Firmware version None	Version column blank or FAIL	packet[4] = 0x15 condition not met	Meter must respond with 0x15 at byte index 4

Production Test Failures

Test	Failure Symptom	Expected	Root Cause	Recovery
Module Supply On (DO)	FAIL returned	"DO output completed!"	BDAQ card not detected; wrong dec_port_id	Reconnect USB-4716; verify device string
LDO Voltage	Below spec	Within LDO spec range	Supply ON failed first; DUT LDO circuit defect	Confirm supply step passed; replace DUT
Super Cap Charge	Voltage too low	Above minimum charge threshold	Charge time too short; defective supercap	Allow longer charge time; inspect supercap
RF RSSI RX	Below threshold	≥ threshold in RF_criteria.json	Antenna disconnected; RF shield not closed	Reseat NIC; close RF shield; re-test
RF RSSI TX	Below threshold	≥ threshold	Faulty NIC module	Replace NIC; re-test
Firmware Version	Returned version ≠ expected	Matches firmware_config.txt	Wrong firmware flashed; DUT not reset after flash	Re-flash; power-cycle DUT; re-test

4.2 3PH LTCT — 3-Phase LTCT Test Jig

Purpose

Tests 3-phase LTCT (Low-Tension Current Transformer) meters via 0xAA-framed serial protocol at 9600 baud. **2 stations.** No DAQ cards used — all measurements via optical/UART serial ports. Tokyo and Zettown factory variants have **different test sequences**.

4.2.1 — 3PH LTCT Tokyo (3PH_LTCT_TOKYO)

Test Sequence — 5 Steps:

Step	Test Name	Action
1	Mapped with COM Port	COM port assigned via serial_config.enc

Step	Test Name	Action
2	Main Firmware Download	Flash via GangProARM DLL
3	Live Version Test	Read firmware version string (0xAA frame)
4	RTC Calibration	Set PPM to 1,000,000; verify <code>compensated_ppm == 0</code> AND <code>ticks == 0</code> AND <code>interval == 1</code>
5	Digital Input Test	Read 4 DI bits; PASS only if returned integer == 15 (all bits set, <code>0b1111</code>)

Firmware Files: - Bootloader: `3PLTCT25_V4.8_bootloader.bin` (default; overridden by `serial_config.enc` key `Bootloader_Firmware`) - Main FW: `3PLTCT25_UP_FW_V1111_0022_main_fw.bin` (default; overridden by `serial_config.enc` key `Main_firmware`) - Merged output: `firmware_flash_files/merged_firmware.bin` (bootloader @ 0x0000, main FW @ 0x2800) - Gang config: `firmware_flash_files\5_6_3Ph_GangConfig.cfg` (from `serial_config.enc` key `Gang_Programming_Code_File`)

Excel Column Schema (`YYYY-MM-DD_3_Phase_RTC_TEST_RESULT.xlsx`) — 15 Columns:

#	Column	#	Column
1	Test Session	9	RTC Calibration Result
2	Station	10	RTC Calibration Value
3	PCBA Serial Number	11	Digital Input Result
4	Test Start Time	12	Digital Input Value
5	Test End Time	13	Overall Status
6	FW Flash Result	14	Total Test Time (s)
7	Live Version Result	15	Timestamp
8	Live Version Value		

4.2.2 — 3PH LTCT Zettown (`3PH_LTCT_ZETTOWN`)

Test Sequence — 15 Steps:

Step	Test Name	Step	Test Name
1	Mapped with COM Port	9	MD Reset Button Test
2	Test Firmware Download	10	Magnet Status Test
3	Memory Test	11	Push Button 1 Test
4	Live Version Test	12	Push Button 2 Test
5	NIC Status Test	13	RX/TX Status Test
6	3P Current Test	14	eFuse Mode Test
7	3P Voltage Test	15	Cover Open Test
8	Digital Input Test		

COM Port Configuration (`config.txt`):

```
SCANNER_COM1=COM28    ← Barcode scanner station 1
SCANNER_COM2=COM29    ← Barcode scanner station 2
OPTICAL_COM1=COM30     ← Optical interface station 1
OPTICAL_COM2=COM27     ← Optical interface station 2
MAX_RETRIES=2
TEST_TIMEOUT=15
RETRY_DELAY=2
```

Firmware Files: - Flash FW: `Test_FW_LTCT_1700_v004.txt` (from `CodeFiles/Firmware_files.txt` key `Flash_firmware`) - Gang config: `ConfigFiles/3Ph_GangConfig.cfg` (from `firmwareflash_config.json`)

Excel Column Schema — 38 Columns:

#	Column	#	Column	#	Column
1	Test Session	14	3P Voltage R (V)	27	Push Button 1 Value
2	Station	15	3P Voltage Y (V)	28	Push Button 2 Result
3	PCBA Serial Number	16	3P Voltage B (V)	29	Push Button 2 Value
4	Test Start Time	17	Digital Input Result	30	RX/TX Status Result
5	Test End Time	18	Digital Input Value	31	RX/TX Status Value
6	FW Flash Result	19	MD Reset Button Result	32	eFuse Mode Result
7	Live Version Result	20	MD Reset Button Value	33	eFuse Mode Value
8	Live Version Value	21	Magnet Status Result	34	Cover Open Result
9	NIC Status Result	22	Magnet Status Value	35	Cover Open Value
10	NIC Status Value	23	Push Button 1 Result	36	Overall Status
11	Memory EEPROM Result	24	Push Button 1 Value	37	Total Test Time (s)
12	Memory Flash Result	25	Push Button 2 Result	38	Timestamp
13	3P Current R/Y/B/N (A) × 4 cols	26	Push Button 2 Value		

Result files: `Test Result Excel File/Test_Result_YYYY-MM-DD/{YYYY-MM-DD}_3_Phase_TEST_RESULT.xlsx` and `{YYYY-MM-DD}_DETAILED_TEST_LOG.xlsx`

Common Errors (Both LTCT Variants)

Error Message	Possible Cause	Impact	Resolution
 <code>Invalid start byte: XX, clearing packet</code>	Cable noise; DUT not ready; wrong COM port	Test hangs	Power-cycle DUT; verify cable; check COM port
 <code>Inter-byte timeout after 3.0s, partial packet</code>	DUT firmware crashed mid-response	Partial data; test fails	Check DUT power; inspect connector pins
		Station FAIL	

Error Message	Possible Cause	Impact	Resolution
<code>Station X: Process timed out after Ys</code>	Meter DUT unresponsive; subprocess hung		<code>ProcessManager</code> auto-kills; check DUT power
<code>Connected adapters: 0</code>	GangProARM FPA USB not detected	Flash cannot proceed	Check USB cable to FPA; verify <code>FPA-setup.ini</code>

Custom Serial Protocol (0xAA Frame)

Byte Index	Field	Notes
0	Start byte	Always <code>0xAA</code>
1–8	Header fields	Device address, command type
9	Port ID	Station/port identifier
12+	Payload	Variable; decoded per test
Last 2	CRC	Packet integrity check

Expected RX frame sizes: 26 bytes (voltage), 29 bytes (current), 17 bytes (version), 15 bytes (magnet).

Firmware Flash Issues (GangProARM)

Error Code	Name	Cause	Resolution
522	<code>STATUS_JTAG_INIT_ERROR</code>	JTAG probe not contacting DUT	Check pogo pins; verify socket seating
523	<code>STATUS_JTAG_COMM_ERROR</code>	JTAG bus communication failure	Check USB cable to FPA; verify FPA power LED
526	<code>STATUS_FW_VERIFICATION_ERROR</code>	Flash content mismatch after write	Re-flash; verify firmware binary integrity
535	<code>STATUS_OPEN_FILE_ERROR</code>	Firmware <code>.bin</code> file not found	Verify <code>firmware_flash_files/</code> contains correct binaries
551	<code>STATUS_LOCKED_DEVICE</code>	OTP bits locked	Escalate to hardware team
549	<code>STATUS_ERASE_SEGMENT_FAILED</code>	Flash erase failed	Poor JTAG contact; unstable PSU

Firmware Merge Logic (`FirmwareFlash.py`): - Bootloader at offset `0x0000` , main firmware at offset `0x2800` - Full flash image: `0x40000` (256 KB) filled with `0xFF` - Merged output: `firmware_flash_files/merged_firmware.bin`

Production Test Failures (LTCT)

Test	Failure	Expected	Root Cause	Recovery
Voltage R/Y/B	Out-of-range	230 ±15% nominal	Reference voltage not applied; CT ratio wrong	Check applied voltage source; verify CT configuration
		Non-zero with load applied		

Test	Failure	Expected	Root Cause	Recovery
Current R/Y/B/N	Value = 0 with load		CT not clamped; wrong phase connection	Verify CT connection per phase
Live firmware version	Returns <code>None</code>	<code>version_XXXX_XXXX</code>	DUT firmware not running	Power-cycle DUT; wait 10s before version check
RTC calibration	Offset > tolerance	<code>compensated_ppm = 0</code> , <code>ticks = 0</code> , <code>interval = 1</code>	DUT crystal defective	Fail DUT; escalate to hardware
Digital Input	Value ≠ 15	<code>0b1111</code> (15) — all 4 DI bits set	Jig DI connections; DUT digital input circuit	Verify all 4 DI signals are applied
Memory test	FAIL	PASS	EEPROM write-verify mismatch	Check DUT power stability

4.3 3PH WC — 3-Phase Whole-Current Test Jig

Purpose

Tests 3-phase whole-current (direct-connect) electricity meters. **2 stations**. No DAQ cards — all measurements via UART/optical serial. Tokyo and Zettown variants have different test sequences and Excel schemas.

4.3.1 — 3PH WC Tokyo (`3PH_WC_TOKYO`)

Test Sequence — 4 Steps:

Step	Test Name	Action
1	Mapped with COM Port	COM port assigned via <code>serial_config.enc</code>
2	Main Firmware Download	Flash via GangProARM DLL
3	Live Version Test	Read firmware version string (0xAA frame)
4	RTC Calibration	Set PPM to 1,000,000; verify pass conditions

Firmware Files: - Bootloader: `3PLTCT25_V4.9_bootloader.bin` (actual file in repo; hardcoded fallback: `3PLTCT25_V4.8_bootloader.bin`) - Main FW: `3PLTCT26_UP_FW_V1111_0032_main_fw.bin` (actual; hardcoded fallback: `3PLTCT25_UP_FW_V1111_0022_main_fw.bin`) - Both runtime names loaded from `serial_config.enc` keys `Bootloader_Firmware` and `Main_firmware` - Gang config: `firmware_flash_files\5_6_3Ph_GangConfig.cfg` (from `serial_config.enc` key `Gang_Programming_Code_File`)

Excel Column Schema (`YYYY-MM-DD_3_Phase_RTC_TEST_RESULT.xlsx`) — 13 Columns:

#	Column	#	Column
1	Test Session	8	Live Version Value
2	Station	9	RTC Calibration Result

#	Column	#	Column
3	PCBA Serial Number	10	RTC Calibration Value
4	Test Start Time	11	Overall Status
5	Test End Time	12	Total Test Time (s)
6	FW Flash Result	13	Timestamp
7	Live Version Result		

4.3.2 — 3PH WC Zettown (`3PH_WC_zettown`)

Test Sequence — 14 Steps (same as LTCT Zettown minus Digital Input Test):

Step	Test Name	Step	Test Name
1	Mapped with COM Port	8	MD Reset Button Test
2	Test Firmware Download	9	Magnet Status Test
3	Memory Test	10	Push Button 1 Test
4	Live Version Test	11	Push Button 2 Test
5	NIC Status Test	12	RX/TX Status Test
6	3P Current Test	13	eFuse Mode Test
7	3P Voltage Test	14	Cover Open Test

Note: `Digital Input Test` is **absent** from WC Zettown (present in LTCT Zettown). Tasks diverge after `3P Voltage Test`.

COM Port Configuration (`config.txt`):

```
SCANNER_COM1=COM28
SCANNER_COM2=COM29
OPTICAL_COM1=COM30
OPTICAL_COM2=COM27
```

Firmware Files: - Flash FW: `Test_FW_LTCT_1700_v004.txt` (from `CodeFiles/Firmware_files.txt` key `Flash_firmware`) - Gang config: `ConfigFiles/3Ph_GangConfig.cfg` (from `firmwareflash_config.json`)

Excel Column Schema — 36 Columns (same as LTCT Zettown minus Digital Input rows):

#	Column	#	Column
1-5	Test Session, Station, PCBA SN, Start/End Time	19	Magnet Status Result
6	FW Flash Result	20	Magnet Status Value
7-8	Live Version Result/Value	21-22	Push Button 1 Result/Value
9-10	NIC Status Result/Value	23-24	Push Button 2 Result/Value

#	Column	#	Column
11	Memory EEPROM Result	25–26	RX/TX Status Result/Value
12	Memory Flash Result	27–28	eFuse Mode Result/Value
13–16	3P Current R/Y/B/N (A)	29–30	Cover Open Result/Value
17–19	3P Voltage R/Y/B (V)	31	Overall Status
—	(no Digital Input)	32	Total Test Time (s)
18	MD Reset Button Result	33	Timestamp

Result files: `Test Result Excel File/Test_Result_YYYY-MM-DD/{YYYY-MM-DD}_3_Phase_TEST_RESULT.xlsx` and `{YYYY-MM-DD}_DETAILED_TEST_LOG.xlsx`

Common Errors (Both WC Variants)

Error	Cause	Resolution
Bootloader or Main firmware not found	<code>serial_config.enc</code> missing or fallback filenames don't match actual files	Verify <code>serial_config.enc</code> ; check firmware files match configured names
Connected adapters: 0	GangProARM FPA USB not detected	Check USB cable; verify <code>FPA-setup.ini</code> path
Incorrect voltage readings	Whole-current reference calibration not applied	Apply reference standard before test
Invalid start byte	Cable noise; DUT not ready	Power-cycle DUT; verify cable

Production Test Failures (WC)

Test	Failure	Expected	Root Cause	Recovery
Voltage (L1/L2/L3)	Out-of-range	230 V $\pm$ 5%	Reference not applied; wrong phase	Verify reference calibrator
Current	0 reading	Load current applied	Direct wire not connected	Check wiring harness
Firmware flash	Any slot failed	All #1–#8 OK	FPA socket not seating	Re-seat DUT; clean spring contacts
RTC calibration	Offset > tolerance	Pass conditions met	DUT crystal defective	Fail DUT; escalate to hardware

4.4 IMG Stage 1 — Firmware Flash & Provisioning Jig

Purpose

IMG (Intelligent Metering Gateway) Stage 1 jig performs firmware flash (GangProARM), optical COM port test, DAQ-controlled power sequencing (USB-4716 DO), analog voltage measurement (USB-4704 AI), and firmware version check. **4 stations**. Requires `pywin32`. Two variants differ in DAQ channel layout and DO values.

Test Sequence (Both Stage 1 Variants — 16 Steps)

Step	Test Name	DO/AI	Action
1	Mapped with COM Port	—	COM port / DEC card port assigned
2	BOOT Mode Initialization	DO	Relay to initialize boot mode
3	BOOT MODE ON	DO	Enable boot mode power
4	Buck Output	AI read	Measure Buck converter voltage
5	LDO Output	AI read	Measure LDO rail voltage
6	1V8	AI read	Measure 1.8 V rail
7	IMG SINK Programming	FW flash	Flash sink firmware via GangProARM DLL
8	N58 Programming Initialization	Stub	<code>is_execute=False</code> — returns PASS without running
9	Super Cap Output	AI read	Measure supercap charge voltage
10	BOOT MODE OFF	DO	De-energize boot mode
11	MODULE SUPPLY ON	DO	Enable module supply
12	Firmware Version Check	Serial	Read GW Core/Main/Sink/Modem/Meter Lib versions
13	MODULE SUPPLY OFF	DO	Power down module supply
14	SUPERCAP DISCHARGE	DO	Active discharge path ON
15	SUPERCAP DISCHARGED VOLTAGE	AI read	Measure supercap after discharge
16	Module Supply Disconnected	DO	Final disconnect state

Note: N58 Programming Initialization (step 8) is a no-op stub — it always returns PASS immediately without executing any programming script.

Startup Flow

- `SmartManufacturingSuite.py` starts; `BASE_DIR` set to executable directory
- `serial_config.enc` decrypted (password: `Polaris@1234`, Fernet/PBKDF2)
- Advantech BDAQ device enumerated — USB-4716 (DO) and USB-4704 (AI)
- 16CH initial DO state: `0x07`; 8CH initial DO state: `0x00`
- Station slots displayed (4 stations)
- Operator loads DUT → clicks Start

DAQ Digital Output Map — Stage 1 16CH DEC Card (USB-4716)

DO Hex Value	Meaning
0x03	BOOT Mode Initialization
0x02	BOOT MODE ON
0x07	Supply OFF / Module Disconnected / Discharge state
0x06	MODULE SUPPLY ON
0x05	SUPERCAP DISCHARGE

DAQ Digital Output Map — Stage 1 8CH DEC Card (USB-4704)

Source: `SmartManufacturingSuite.py` branch `IMG_STAGE_1_DEC_2_CH_8`, lines 1720–1786 (`--inputVal` args passed to `staticDO.py`)

DO Hex Value	Test Step	Meaning
0x01	BOOT Mode Initialization	Initialize module boot mode
0x05	BOOT MODE ON	Enable boot mode power
0x00	BOOT MODE OFF	De-energize boot mode
0x04	MODULE SUPPLY ON	Turn on module supply after programming
0x00	MODULE SUPPLY OFF	Power down module
0x02	SUPERCAP DISCHARGE	Active supercap discharge

Important: Stage 1 8CH DO values (`0x01` , `0x05` , `0x00` , `0x04` , `0x02`) are **different** from Stage 1 16CH (`0x03` , `0x02` , `0x07` , `0x06` , `0x05`). Deploying the wrong branch on the wrong card type will cause incorrect power sequencing.

DAQ AI Channel Map — Stage 1 (16CH DEC Card) — `IMG_STAGE_1_DEC_1_CH_16`

Station	Buck Voltage	LDO Output	1V8 Voltage	SuperCap
1	CH4	CH8	CH12	CH0
2	CH5	CH9	CH13	CH1
3	CH6	CH10	CH14	CH2
4	CH7	CH11	CH15	CH3

DAQ AI Channel Map — Stage 1 (8CH DEC Card) — `IMG_STAGE_1_DEC_2_CH_8`

Source: channel formula `(station_num - 1) × 4 + offset`, lines 2101–2142

Station	Buck Output	LDO Output	SuperCap Voltage	1V8 Voltage	SuperCap Discharged
1	CH0	CH1	CH2	CH3	CH2
2	CH4	CH5	CH6	CH7	CH6
3	CH8	CH9	CH10	CH11	CH10
4	CH12	CH13	CH14	CH15	CH14

All 16 AI channels used (4 measurements × 4 stations). SuperCap Discharged voltage is read on the same channel as SuperCap Voltage ($(station_num-1) \times 4 + 2$).

Excel Column Schema (`YYYY-MM-DD_IMG_TEST_RESULT.xlsx`) — 38 Columns (Both Stage 1 Variants)

#	Column	#	Column
1	Test Session	20	Module Supply On
2	Station	21	GW Core Ver
3	PCBA Serial Number	22	GW Core Ver Status
4	Test Start Time	23	GW Main Ver
5	Test End Time	24	GW Main Ver Status
6	Mapped with COM Port	25	GW Sink App Ver
7	BOOT Mode Initialization	26	GW Sink App Ver Status
8	BOOT mode on	27	GW Modem Firmware Ver
9	Buck Output	28	GW Modem Firmware Ver Status
10	Buck Output Status	29	GW Meter Lib Ver
11	LDO Output	30	GW Meter Lib Ver Status
12	LDO Output Status	31	Version Check
13	1V8	32	Module Supply Off
14	1V8 Status	33	SUPERCAP DISCHARGE
15	IMG SINK Programming	34	SUPERCAP DISCHARGED VOLTAGE
16	N58 Programming Initialization	35	SUPERCAP DISCHARGED VOLTAGE Status
17	Super Cap Output / Super Cap Voltage	36	Overall Status
18	Super Cap Output Status	37	Total Test Time (s)
19	BOOT Mode Off	38	Timestamp

Note: 16CH branch Excel header uses `'Super Cap Output'` and `'Super Cap Output Status'` for columns 17–18; 8CH branch uses `'Super Cap Voltage'` and `'Super Cap Voltage Status'`.

Result file: `Test Result Excel File/Test_Result_YYYY-MM-DD/{YYYY-MM-DD}_IMG_TEST_RESULT.xlsx`

Common Errors (IMG Stage 1)

Error	Possible Cause	Impact	Resolution
<code>BDAQApi</code> device not found	USB-4716 or USB-4704 not plugged in	All power/ analog tests fail	Plug in DAQ USB; install Advantech DAQ driver
<code>InstantAiCtrl</code> read returns -1	AI channel index out-of-range (16CH bug: <code>channelCount=8</code> but reading CH8+)	LDO and 1V8 always fail	Update <code>channelCount=16</code> in <code>InstantAI.py</code> (16CH branch bug)
Fernet decryption error on <code>serial_config.enc</code>	Wrong password or corrupted file	COM ports not loaded	Use password <code>Polaris@1234</code> ; regenerate <code>serial_config.enc</code>
<code>STATUS_JTAG_INIT_ERROR</code>	Pogo pins not contacting DUT in 8-socket fixture	Flash fails	Inspect and clean pogo pins; verify DUT orientation
Optical COM test fails	Optical probe not seated on DUT	Test FAIL	Reseat optical probe; verify DUT alignment
LDO voltage = 0	AI channel range bug (16CH) or LDO not enabled	LDO test FAIL	Check 16CH <code>channelCount</code> ; verify MODULE SUPPLY ON DO applied

Configuration Issues

Issue	Cause	Resolution
<code>serial_config.enc</code> not found	File not deployed	Copy from repository branch root to executable folder
Firmware binary mismatch	Wrong firmware files in <code>firmware_flash_files/</code>	Verify filenames in <code>serial_config.enc</code> key <code>Flash_Firmware</code> match files present
Wrong channel readings	8CH vs 16CH branch mismatch	Use <code>IMG_STAGE_1_DEC_1_CH_16</code> for 16CH DAQ jig; <code>IMG_STAGE_1_DEC_2_CH_8</code> for 8CH

4.5 IMG Stage 2 — Boost & Supercap Power Test Jig (8CH / 16CH)

Purpose

IMG Stage 2 performs purely electrical power-path testing of the IMG module using the Advantech BDAQ DAQ cards. It does **not** perform firmware flash, RF testing, or provisioning. The test verifies: - Module power-up/ boot-mode sequencing via USB-4716 Digital Output - Boost converter output voltage via USB-4704 Analog Input - Supercapacitor charge voltage via USB-4704 Analog Input - Supercapacitor discharge voltage via USB-4704 Analog Input (after active discharge)

Results are written to `YYYY-MM-DD_IMG_TEST_RESULT.xlsx` (22 columns). Uses `pywin32` for Windows COM automation.

Test Sequence (Both Stage 2 Variants — 10 Steps)

Step	Test Name	DO/AI	Action
1	Mapped with COM Port	—	Assigns USB-4716 DEC card port ID to station
2	BOOT Mode Initialization	DO → USB-4716	Initializes module boot mode relay state
3	BOOT MODE ON	DO → USB-4716	Enables boot mode power (110 s wait after this step)
4	Boost Output	AI read → USB-4704	Reads boost converter output voltage
5	Super Cap Voltage	AI read → USB-4704	Reads supercap voltage during charge (simultaneous with step 4)
6	BOOT MODE OFF	DO → USB-4716	Disables boot mode
7	SUPERCAP DISCHARGE	DO → USB-4716	Actively discharges supercapacitor (50 s wait after this step)
8	DISCHARGE OUTPUT	AI read → USB-4704	Reads discharge output voltage
9	Discharge Super Cap Voltage	AI read → USB-4704	Reads supercap voltage during discharge (simultaneous with step 8)
10	MODULE SUPPLY OFF	DO → USB-4716	Powers down module

DO values differ between 8CH and 16CH variants — see tables below. AI channel formulas also differ.

DAQ Digital Output Map — Stage 2 8CH DEC Card (USB-4704) — `IMG_STAGE_2_1_DEC_8_CH`

Source: `SmartManufacturingSuite.py` lines 1229–1248, `args_var.append(...)` calls to `staticDO.run_staticDO`

DO Hex Value	Test Step	Meaning
<code>0x01</code>	BOOT Mode Initialization	Initialize module boot mode
<code>0x05</code>	BOOT MODE ON	Enable boot mode power supply
<code>0x00</code>	BOOT MODE OFF	De-energize boot mode
<code>0x02</code>	SUPERCAP DISCHARGE	Active supercap discharge path ON
<code>0x00</code>	MODULE SUPPLY OFF	Final power-down

DAQ Digital Output Map — Stage 2 16CH DEC Card (USB-4716) — `IMG_STAGE_2_1_DEC_16_CH`

Source: `SmartManufacturingSuite.py` lines 1229–1248 of 16CH branch — different values from 8CH branch

DO Hex Value	Test Step	Meaning
<code>0x03</code>	BOOT Mode Initialization	Initialize module boot mode

DO Hex Value	Test Step	Meaning
0x02	BOOT MODE ON	Enable boot mode power supply
0x07	BOOT MODE OFF	De-energize boot mode
0x05	SUPERCAP DISCHARGE	Active supercap discharge path ON
0x07	MODULE SUPPLY OFF	Final power-down

Critical: Stage 2 8CH and 16CH cards use **different DO values** for the same test steps. Using the wrong branch on the wrong card will apply wrong voltage states to the DUT. Always verify the deployed branch matches the physical card installed.

DAQ AI Channel Map — Stage 2 8CH DEC Card — IMG_STAGE_2_1_DEC_8_CH

Source: channel formula $(\text{station_num} - 1) \times 2 + \text{offset}$, lines 1358–1408

Station	Super Cap Voltage	Boost Output	Discharge SuperCap	Discharge Output
1	CH0	CH1	CH0	CH1
2	CH2	CH3	CH2	CH3
3	CH4	CH5	CH4	CH5
4	CH6	CH7	CH6	CH7

Discharge channels share the same AI index as charge channels — the measurement is taken after DO 0x02 discharge state is applied.

DAQ AI Channel Map — Stage 2 16CH DEC Card — IMG_STAGE_2_1_DEC_16_CH

Source: channel formula $(\text{station_num} - 1) + 0$ (SuperCap) and $(\text{station_num} - 1) + 4$ (Boost), lines 1354–1392

Station	Super Cap Voltage	Boost Output	Discharge SuperCap	Discharge Output
1	CH0	CH4	CH0	CH4
2	CH1	CH5	CH1	CH5
3	CH2	CH6	CH2	CH6
4	CH3	CH7	CH3	CH7

Excel Column Schema (`YYYY-MM-DD_IMG_TEST_RESULT.xlsx`)

#	Column Header (exact)	Type	Description
1	Test Session	String	Session ID
2	Station	Int	Jig station number (1–4)
3	PCBA Serial Number	String	Scanned barcode
4	Test Start Time	Timestamp	Test start
5	Test End Time	Timestamp	Test end
6	Mapped with COM Port	PASS/FAIL	DEC card port assignment result
7	BOOT Mode Initialization	PASS/FAIL	DO <code>0x01</code> write result
8	BOOT MODE ON	PASS/FAIL	DO <code>0x05</code> write result
9	Boost Output	Float (V)	Measured boost voltage value
10	Boost Output Status	PASS/FAIL	Boost voltage within spec?
11	Super Cap Voltage	Float (V)	Measured supercap charge voltage
12	Super Cap Voltage Status	PASS/FAIL	Supercap charge voltage within spec?
13	BOOT MODE OFF	PASS/FAIL	DO <code>0x00</code> write result
14	SUPERCAP DISCHARGE	PASS/FAIL	DO <code>0x02</code> write result
15	Discharge Output	Float (V)	Measured discharge output voltage
16	Discharge Output Status	PASS/FAIL	Discharge output within spec?
17	Discharge Super Cap Voltage	Float (V)	Measured supercap discharge voltage
18	Discharge Super Cap Voltage Status	PASS/FAIL	Supercap discharge within spec?
19	MODULE SUPPLY OFF	PASS/FAIL	DO <code>0x00</code> final power-down result
20	Overall Status	PASS/FAIL	All steps passed?
21	Total Test Time (s)	Float	Duration in seconds
22	Timestamp	Timestamp	Excel write time

Common Errors (IMG Stage 2)

Error	Possible Cause	Impact	Resolution
<code>BadZipFile</code> on Excel open	Excel corrupted by prior power loss	Result row lost for session	App auto-renames to <code>.corrupted</code> ; new file auto-created; recover data from <code>.corrupted</code>
<code>pythoncom</code> / <code>win32gui</code> import error	<code>pywin32</code> not installed or post-install not run	App won't start	<code>pip install pywin32</code> then run <code>python Scripts/pywin32_postinstall.py -install</code> as Admin
BDAQ device not found	USB-4716 or USB-4704 not detected	All DO/AI steps fail	Plug in DAQ USB; install Advantech driver; verify <code>USB-4716,BID#0</code> device string

Error	Possible Cause	Impact	Resolution
DO write returns error code	USB-4716 fault or wrong device description	BOOT/ DISCHARGE steps FAIL	Check <code>dec_port_id</code> set correctly; reconnect USB-4716
Boost voltage = 0	Boot mode not ON (DO step failed); DUT boost circuit defective	Boost Output FAIL	Confirm <code>BOOT MODE ON</code> (DO <code>0x05</code>) completed successfully before AI read
Super Cap voltage not rising	Boost converter not running; charge path open	Super Cap Voltage FAIL	Verify DO <code>0x05</code> applied; check module soldering; measure boost rail with DMM
Discharge voltage not dropping	DO <code>0x02</code> not applied; discharge FET open	Discharge Super Cap FAIL	Verify <code>SUPERCAP DISCHARGE</code> DO <code>0x02</code> status is PASS; check jig discharge circuit
AI reading stuck at same value	USB-4704 channel conflict; DAQ not reading	Wrong voltage readings	Check AI card USB; verify channel index formula $(station-1) \times 2 + 0/1$
Station result FAIL despite voltages in range	Threshold config too tight	Overall Status FAIL	Review <code>min_Boost_voltage</code> , <code>min_super_cap_voltage</code> , <code>max_Dis_super_cap_voltage</code> in config
<code>multiprocessing</code> spawn error	Windows <code>spawn</code> context issue	App crashes on launch	Verify <code>multiprocessing.freeze_support()</code> guard present in <code>__main__</code> entry

Production Test Failures (IMG Stage 2)

Test Step	Failure Symptom	Expected	Root Cause	Recovery
BOOT Mode Initialization (DO <code>0x01</code>)	FAIL status returned	<code>"DO output completed!"</code>	USB-4716 not detected; <code>BioFailed</code> error	Reconnect USB-4716; verify <code>USB-4716,BID#0</code> ; reinstall Advantech driver
BOOT MODE ON (DO <code>0x05</code>)	FAIL status	<code>"DO output completed!"</code>	Same as above	Same as above
Boost Output	Voltage = 0 or below <code>min_Boost_voltage</code>	Boost converter output within spec range	BOOT MODE ON failed first; boost circuit defective	Confirm step 3 passed; replace DUT if boost rail absent
Boost Output Status	FAIL even with non-zero voltage	PASS within configured min/max	Threshold too tight; AI offset	Check voltage spec in config; run AI calibration
Super Cap Voltage	Voltage = 0 or very low	Supercap charges above min threshold	Supercap missing/ defective; charge path broken	Inspect supercap soldering; replace DUT
	FAIL	PASS		Allow adequate charge time; verify threshold

Test Step	Failure Symptom	Expected	Root Cause	Recovery
Super Cap Voltage Status			Charge time too short; threshold mismatch	
BOOT MODE OFF (DO <code>0x00</code>)	FAIL status	"DO output completed!"	USB-4716 USB cable fault	Check USB-4716 cable; reconnect
SUPERCAP DISCHARGE (DO <code>0x02</code>)	FAIL status	"DO output completed!"	DO card fault	Check USB-4716; verify <code>dec_port_id</code>
DISCHARGE OUTPUT	Voltage not dropping	Should be near 0 after discharge	Discharge FET not conducting; DO <code>0x02</code> failed	Verify DO <code>0x02</code> status PASS; check jig discharge FET
Discharge Super Cap Voltage	Voltage > <code>max_Dis_super_cap_voltage</code>	Supercap fully discharged	Discharge time too short; high-capacitance DUT	Extend software discharge wait; check spec
MODULE SUPPLY OFF (DO <code>0x00</code>)	FAIL status	"DO output completed!"	USB-4716 fault	Reconnect USB-4716; replace if failed
Overall Status	FAIL despite all steps looking OK	PASS	One column status FAIL not visible	Inspect all 22 Excel columns for any FAIL value

5. Software Logs Analysis Guide

5.1 Log Locations

Model	Log File Path	Rotation
All	<code><app_dir>/Log_files/Log_file_YYYY-MM-DD/YYYY-MM-DD_program_log.txt</code>	Daily (DynamicLogger)
IMG STG2	<code><app_dir>/Log_files/Log_file_YYYY-MM-DD/action_log.txt</code>	Daily
IMG provision	<code>scripts/tests/IMG_Provisioning_Testing_Scripts/Detailed Test Result/DD_MM_YYYY_<serial>.txt</code>	Per DUT

5.2 Log Format

```

2026-06-15 09:23:41,123 - DEBUG - Station 2: Starting process for voltage_test with 15s timeout
2026-06-15 09:23:42,456 - INFO - Station 2: Process completed normally in 1.33s
2026-06-15 09:23:42,460 - ERROR - Station 3: Process timed out after 15 seconds, attempting forced termination

```

5.3 Log Severity Levels

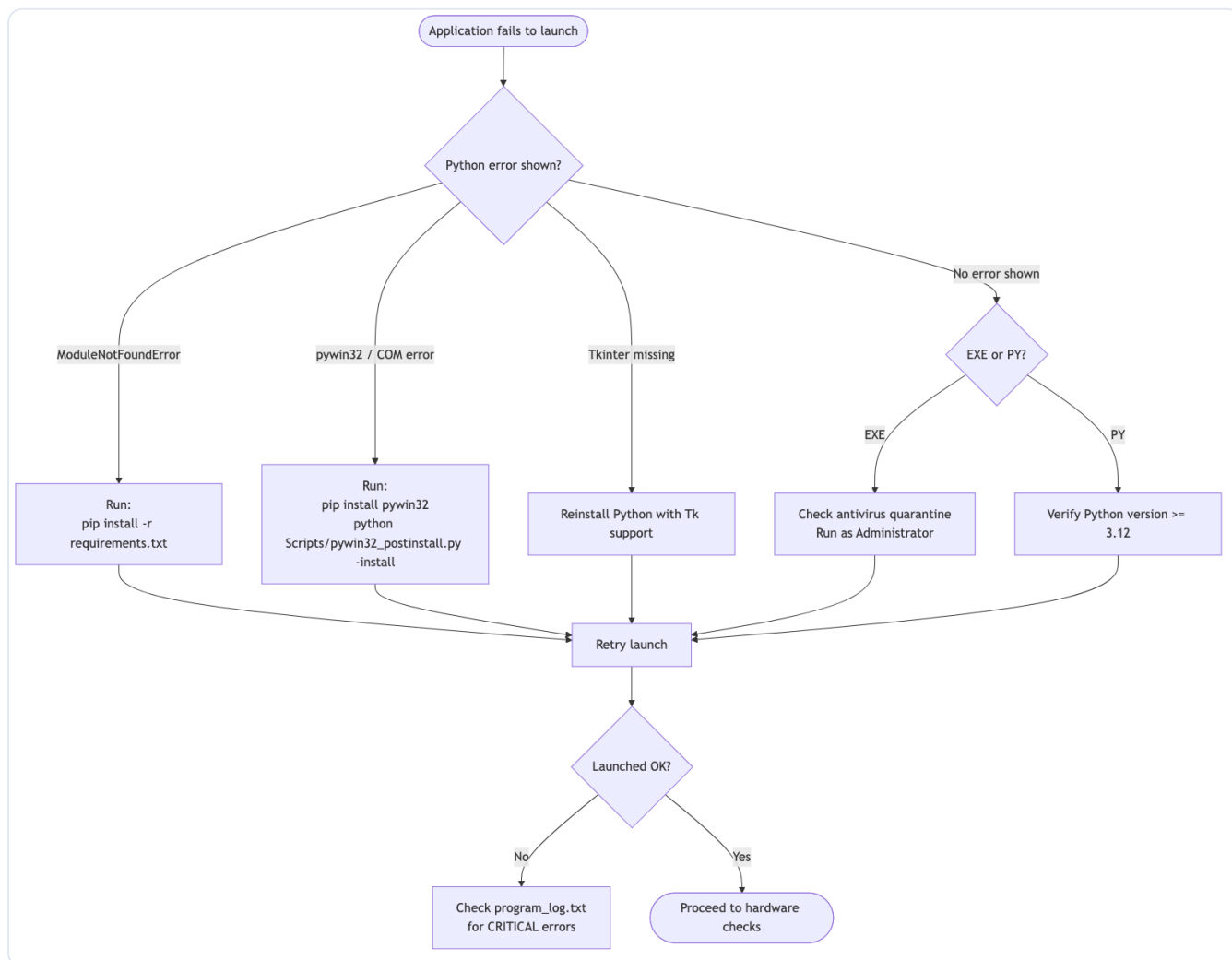
Level	Meaning	Action Required
DEBUG	Normal flow, byte-level trace	None
INFO	Milestones (version read, COM open)	None
WARNING	Non-fatal anomaly (thread stop, retry)	Monitor; investigate if frequent
ERROR	Test failure, timeout, exception	Investigate immediately
CRITICAL	App-level exception (not currently used)	Restart app; escalate

5.4 Critical Log Keywords → Actions

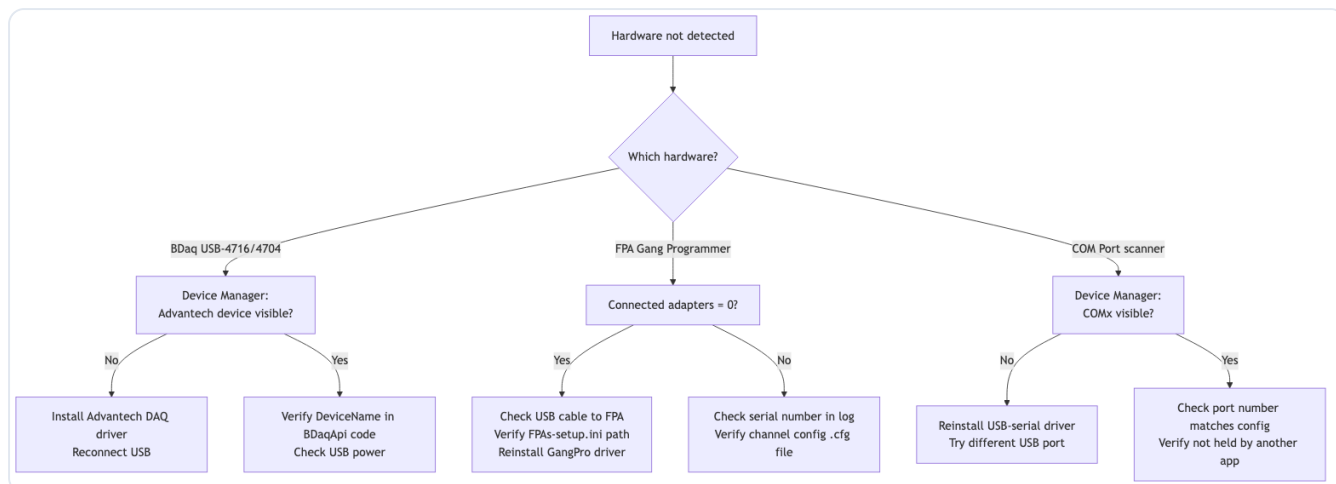
Log Message / Keyword	Meaning	Immediate Action
Process timed out after	Subprocess did not complete	Check DUT power; kill orphan processes
Force killing process tree	Timeout recovery triggered	Normal behavior; investigate if frequent
Failed to open serial port after 3 retries	COM port unavailable	Check USB adapter; Device Manager
Invalid start byte	Frame synchronization lost	Check cables; DUT power
Inter-byte timeout	Partial packet received	Inspect DUT firmware; cable signal
Connected adapters: 0	No FPA detected	USB cable to gang programmer
STATUS_JTAG_COMM_ERROR	Flash adapter JTAG failure	JTAG connector / pogo pins
Bootloader or Main firmware not found	Config/file mismatch	Verify firmwareflash_config.json
Switched log file	Normal daily rotation	No action
Creating new Excel file	Corruption recovery	Check prior data in .corrupted file
Failed to decode	Protocol decode error	Check firmware version; packet format
Complete packet received	Successful frame	No action

6. Diagnostic Flowcharts

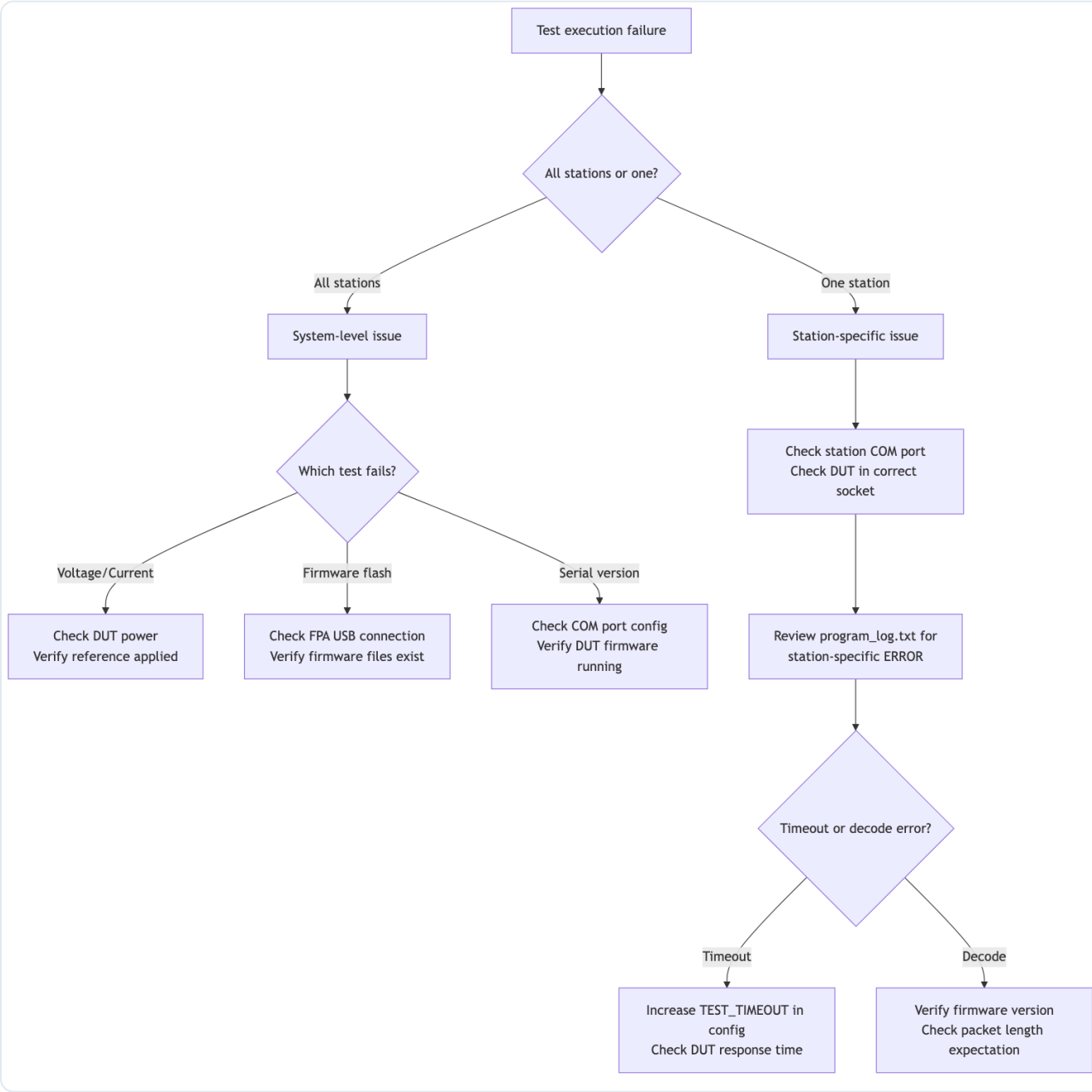
6.1 Application Not Launching



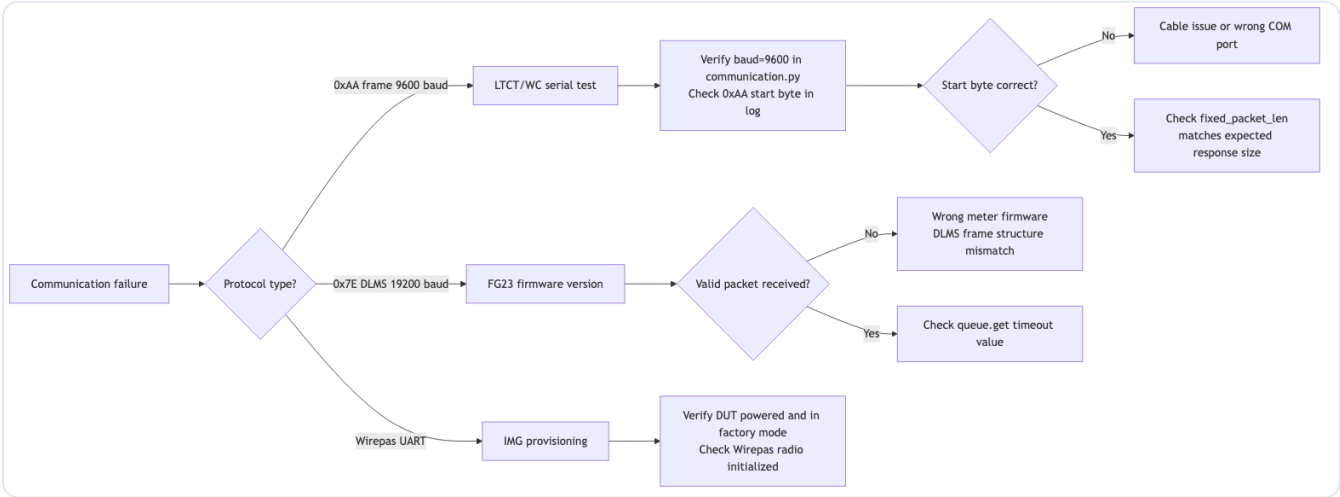
6.2 Hardware Not Detected



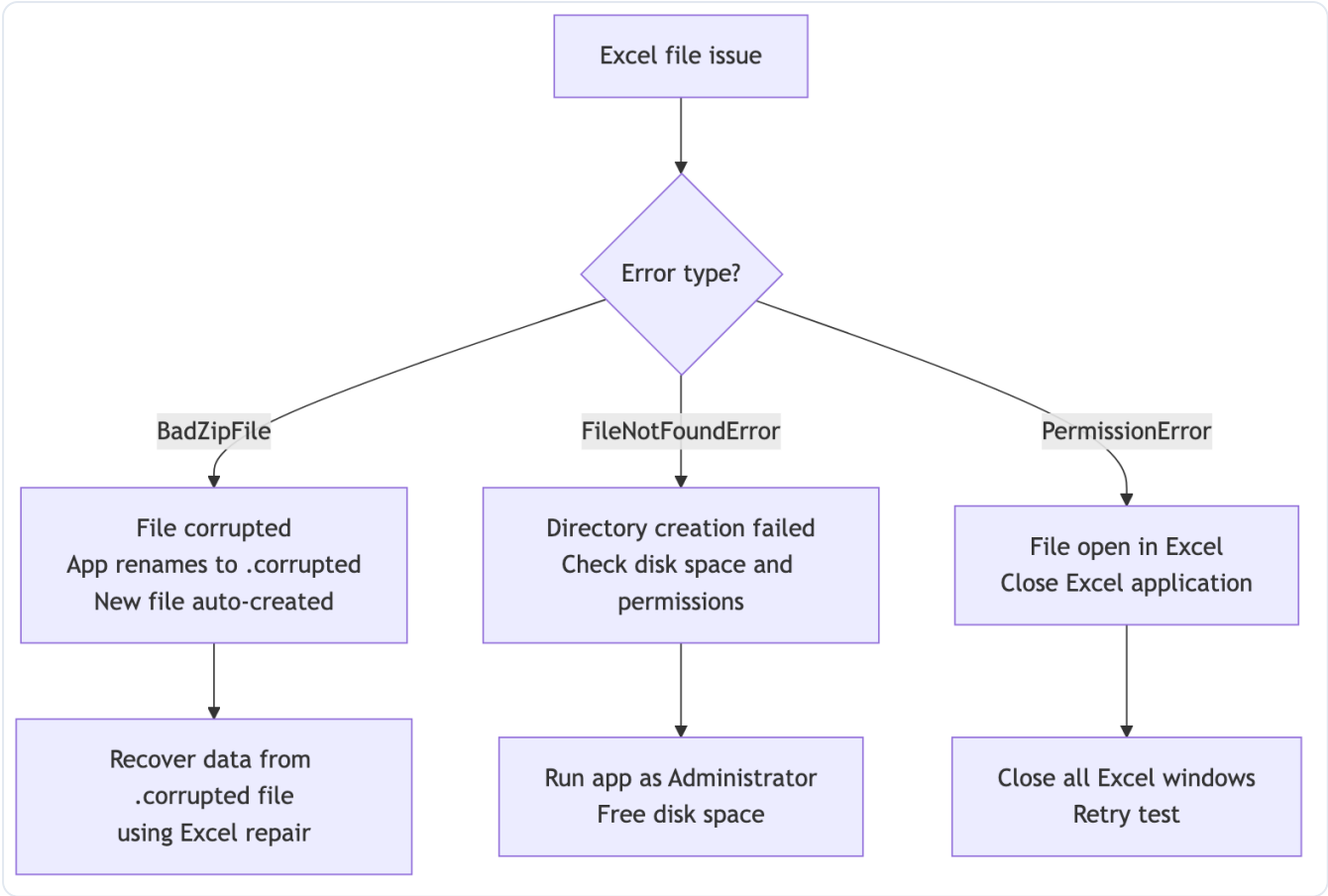
6.3 Test Execution Failure



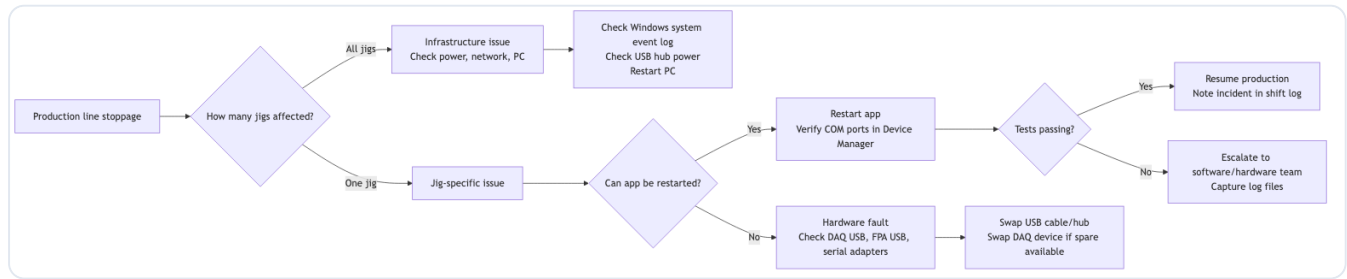
6.4 Communication Failure



6.5 Database / File Failure



6.6 Production Line Failure



7. Root Cause Analysis (RCA) Matrix

Issue	Symptoms	Root Cause	Verification	Resolution
All stations show timeout	<code>Process timed out</code> in all logs	System load too high; Python GIL contention	CPU usage > 90% in Task Manager	Close background apps; reduce parallel thread count
Single station always fails	One station ERROR in log	Hardware fault (cable, socket, DUT) at that position	Swap DUT to another station; re-test	Replace cable/socket for that station
Firmware flash: all sockets fail	<code>#1 – #8</code> all <code>failed</code>	FPA USB disconnected; <code>.ini</code> file path wrong	Check <code>Connected adapters</code> log line	Reconnect FPA; verify <code>FPA-setup.ini</code> path
Firmware flash: 1–2 sockets fail	Some <code>OK</code> , some <code>failed</code>	Pogo pin contact failure at specific socket	Re-seat DUT; examine pogo pins	Clean/replace pogo pins; re-seat DUT
Excel corrupted every day	Frequent <code>.corrupted</code> rename	PC power loss or forced shutdown mid-write	Check Windows Event Log for shutdown	Enable UPS; implement atomic Excel write (write to temp file, rename)
RSSI always below threshold	RF FAIL on all DUTs	Antenna not connected on jig reference side; wrong RF criteria JSON	Measure RSSI with known-good DUT	Reconnect reference antenna; calibrate RSSI threshold
Supercap never charges to threshold	Supercap Status always FAIL	DO <code>0x06</code> not applied; PSU current limit	Probe DO output on oscilloscope; measure supply rail	Verify DO write; increase PSU current limit
COM port number changes on reboot	All COM tests fail after reboot	USB-serial adapter auto-reassigned different COM	Check Device Manager COM numbers	Assign fixed COM numbers in Device Manager (port properties)
<code>pywin32</code> COM error	Exception on <code>pythoncom</code> init	pywin32 post-install script not run	<code>import win32api</code> in Python REPL	Run <code>python Scripts/pywin32_postinstall.py -install</code> as Administrator

Issue	Symptoms	Root Cause	Verification	Resolution
Version check inconsistent	Some pass, some fail same FW	RTC not stabilized; DLMS response timing	Add 10s wait after power-on	Increase power-on settle time before version read

8. Recovery Procedures

8.1 Software Crash

1. Do **not** power-cycle jig hardware immediately
2. Open Task Manager → kill any orphan `python.exe` or `SmartManufacturingSuite.exe`
3. Check `Log_files/Log_file_YYYY-MM-DD/YYYY-MM-DD_program_log.txt` for last error
4. Note affected station(s) and serial numbers
5. Re-launch application
6. Verify COM ports visible in Device Manager before restarting tests
7. Re-test DUTs that were in-progress at crash time (results may be incomplete)

8.2 Unexpected Shutdown / Power Loss

1. On restart, application auto-creates new Excel file if `.xlsx` is corrupted
2. Locate `<date>_*.xlsx.corrupted` — attempt recovery via Excel → File → Open and Repair
3. Verify log files are intact for audit trail
4. Re-test all DUTs that were in the jig at time of shutdown

8.3 Database / Excel Corruption

1. Navigate to `Test Result Excel File/Test_Result_YYYY-MM-DD/`
2. Identify `.corrupted` file
3. Copy to safe location
4. Open with Excel → select "Open and Repair" → extract data
5. Application will auto-create fresh `.xlsx` on next run
6. Manually transfer recovered rows to new file if needed

8.4 Configuration Corruption (`serial_config.enc`)

1. Encrypted file is regenerated from source — obtain fresh copy from repository (correct branch)
2. Decryption password: `Polaris@1234`
3. If regeneration needed, use the same Fernet/PBKDF2 key derivation used in `SmartManufacturingSuite.py`
4. For `config.txt` (LTCT Zettown): restore from version control or template: `SCANNER_COM1=COM28
SCANNER_COM2=COM29 OPTICAL_COM1=COM30 OPTICAL_COM2=COM27 MAX_RETRIES=2 TEST_TIMEOUT=15 RETRY_DELAY=2`
5. Update COM port numbers to match current Device Manager assignments

8.5 Hardware Communication Loss

1. Unplug all USB-serial adapters and USB-DAQ devices
2. Wait 10 seconds
3. Replug in order: USB-serial adapters first, then DAQ devices
4. Open Device Manager; verify all COM ports and DAQ devices appear
5. If COM numbers changed — update `config.txt` / `serial_config.enc` accordingly
6. Restart application

8.6 Firmware Flash Recovery

1. If flash interrupted mid-sequence: DUT may be in undefined state
2. Do **not** power-cycle DUT while FPA is connected and DLL active
3. Restart application; re-attempt flash from scratch
4. If `STATUS_LOCKED_DEVICE (551)` : DUT requires mass erase — escalate to hardware team
5. If `STATUS_FW_VERIFICATION_ERROR (526)` : re-generate `merged_firmware.bin` by running `merge_bin_to_memory_layout()` manually; verify source `.bin` files are intact

8.7 Production Interruption

1. Log DUT serial numbers currently in jig
2. Record last passing test step from log file
3. Remove DUTs; label as "interrupted — re-test required"
4. Restore application and hardware
5. Re-test all removed DUTs from beginning of test sequence
6. Do not accept partial results for DUTs interrupted during supercapacitor charge/discharge cycle

9. Preventive Maintenance Checklist

Daily Checklist

#	Task	Method	Pass Criterion
1	Verify application launches without errors	Start <code>SmartManufacturingSuite.exe</code>	GUI opens; no error dialogs
2	Check all COM ports visible in Device Manager	Open Device Manager	Expected COMx ports listed
3	Run one known-good DUT through full test	Use reference golden sample	All tests PASS
4	Verify log file created for today	Check <code>Log files/Log_file_YYYY-MM-DD/</code>	File exists and has recent timestamps

#	Task	Method	Pass Criterion
5	Inspect pogo pins / jig sockets	Visual inspection	No bent or stuck pins
6	Check USB cable connections (all DAQ, FPA, serial)	Physical inspection	All cables securely connected
7	Verify Excel result file writes correctly	Complete one test cycle	Row appears in <code>.xlsx</code>

Weekly Checklist

#	Task	Method	Pass Criterion
1	Clean pogo pin sockets	IPA and swab	No oxidation or debris
2	Archive previous week's log files	Move to network share / NAS	Files moved; local folder cleared
3	Verify firmware files unchanged	<code>md5sum</code> on firmware binaries	Hash matches reference
4	Check BDAQ device calibration	Run Advantech calibration utility	Within spec
5	Inspect USB hubs / cables for wear	Physical check	No fraying or loose connectors
6	Review error log summary	Grep <code>ERROR</code> in log files	No recurring unexplained errors
7	Verify all GangConfig files present	<code>ls firmware_flash_files/</code>	All expected <code>.cfg</code> , <code>.bin</code> , <code>.ini</code> present

Monthly Checklist

#	Task	Method	Pass Criterion
1	Update Python dependencies	<code>pip install -r requirements.txt --upgrade</code>	No breaking changes (test first)
2	Pull latest branch from repository	<code>git pull</code>	Code up to date
3	Validate RF RSSI thresholds vs process spec	Review <code>RF_criteria.json</code> with NPI	Thresholds match current spec
4	Verify Windows auto-update does not break COM assignments	Test after patch Tuesday	All COM ports stable post-update
5	Backup <code>serial_config.enc</code> and <code>config.txt</code>	Copy to secure network location	Backup exists
6	Check PSU output voltage with calibrated meter	Multimeter on jig supply rails	Within $\pm 1\%$ of nominal
7	Verify pywin32 (IMG Stage 2) functional after Windows updates	Run test; no <code>pythoncom</code> error	Clean startup

Quarterly Checklist

#	Task	Method	Pass Criterion
1	Full jig calibration / certification	Compare against NIST-traceable reference	Calibration certificate issued
2	Replace pogo pins (high-use sockets)	Visual inspection + contact resistance test	< 10 mΩ per pin
3	Validate firmware version compatibility matrix	Test with oldest and newest FW	All pass
4	Review and update troubleshooting document	Engineering review	Document version incremented
5	Perform DAQ hardware self-test	Advantech diagnostics utility	All channels within spec
6	Check and renew antivirus exclusions for jig app folder	AV management console	App folder excluded from scan

10. Version Compatibility Matrix

Software Version	Python Version	Windows Version	Advantech BDaq	GangProARM DLL	Supported Models
V1.0.0.3 (master)	3.12	Win 10/11	USB-4716 + USB-4704	FPAsel.dll	IMG Stage 2 (8-socket card)
V1.0.0.4 (IMG-STG2)	3.12 / 3.13	Win 10/11	USB-4716 + USB-4704	Not used	IMG Stage 2 (8CH + 16CH)
V1.0.0.5 (IMG-STG1)	3.12	Win 10/11	USB-4716 + USB-4704	FPAsel.dll	IMG Stage 1 (16CH)
Current (LTCT/ WC)	3.12	Win 10/11	Not used	FPAsel.dll	3PH LTCT, 3PH WC
Current (FG23)	3.12	Win 10/11	Not used	FPAsel.dll	FG23 NIC RF

Firmware	Compatible Jig Software	Baud Rate	Protocol
3PLTCT26_UP_FW_V1111_0032	LTCT Tokyo/Zettown	9600	0xAA custom frame
3PLTCT25_V4.9 (bootloader)	LTCT, WC	N/A (flash only)	GangProARM JTAG
IMG NIC FW (Wirepas)	IMG STG1, STG2	UART	Wirepas proprietary

11. Deployment & Upgrade Troubleshooting

11.1 Installation Issues

Issue	Cause	Resolution
<code>pip install</code> fails	No internet access on production PC	Pre-build Python wheel packages offline; use <code>pip install --find-links ./wheels -r requirements.txt</code>
<code>pywin32</code> not working after install	Post-install script not run	<code>python Scripts/pywin32_postinstall.py -install</code> as Administrator
BDAQ Python not importable	Advantech SDK not installed	Install Advantech DAQ professional driver; then install BDAQ Python bindings
<code>GangProARM-FPAsel.dll</code> not found	File not in <code>firmware_flash_files/</code>	Copy DLL from reference build; verify <code>ctypes.cdll.LoadLibrary()</code> path
<code>cryptography</code> Fernet error	Wrong password or version mismatch	Regenerate <code>serial_config.enc</code> with current <code>cryptography</code> library

11.2 Upgrade Failures

Issue	Cause	Resolution
New branch code crashes on startup	Dependency version incompatible	Always run <code>pip install -r requirements.txt</code> after branch pull
COM port assignments change post-upgrade	Windows driver update	Reassign fixed COM numbers in Device Manager
Excel column schema changed	New test columns added to new version	Back up and archive old result files; new version creates updated schema
<code>serial_config.enc</code> format changed	Encryption key derivation updated	Regenerate config file from current source

11.3 Rollback Procedure

1. `git checkout <previous-branch-name>` or restore from backup ZIP
2. Run `pip install -r requirements.txt` for previous version
3. Restore `serial_config.enc` and `config.txt` from backup
4. Verify COM ports unchanged
5. Run golden sample validation before resuming production

11.4 Missing Dependencies

```
# Standard install
pip install pyserial openpyxl Pillow crcmod cryptography psutil

# Windows-specific (IMG Stage 2)
pip install pywin32
python Scripts/pywin32_postinstall.py -install

# Verify Advantech BDAQ SDK installed
python -c "from scripts.Automation.BDAQ.InstantDoCtrl import *; print('BDAQ OK')"
```

12. Emergency Troubleshooting Quick Reference

Problem	Immediate Action	Escalation Required?
App won't start	Kill all <code>python.exe</code> ; reinstall dependencies; check antivirus	Only if reinstall fails
All stations timeout	Restart app; check USB hub power	If recurs after restart
COM port lost after reboot	Device Manager → assign fixed COM to adapter → restart app	No
Firmware flash: 0 adapters connected	Reconnect FPA USB → restart app	If adapter LED not on
Firmware flash: all sockets failed	Inspect pogo pins; re-seat DUT; retry	If pins visibly damaged
Excel file corrupted	App auto-recovers; check <code>.corrupted</code> file for data	No (routine recovery)
Supercap never charges	Check PSU rail; verify DO <code>0x06</code> written; replace DUT if PSU OK	If PSU fault confirmed
RF RSSI always fails	Reconnect reference antenna; verify RF criteria JSON thresholds	If antenna/cable OK
<code>pywin32</code> error on launch	Run <code>pywin32_postinstall.py -install</code> as Admin; reboot	No
DUT serial number not scanning	Check scanner USB; verify <code>*T</code> trigger; baud 9600	If scanner hardware faulty
Jig PC frozen / BSOD	Hard reboot; run <code>chkdsk</code> ; verify no hardware driver conflict	If BSOD repeats
Version check always fails	Verify DUT firmware matches expected version; check baud 19200 vs 9600	If new FW version released
Process never terminates	<code>ProcessManager._force_kill_process_tree()</code> auto-kills after timeout; if stuck — Task Manager kill	If kill fails repeatedly
<code>STATUS_LOCKED_DEVICE (551)</code>	Do not retry; label DUT; remove from jig	Yes — hardware team needed

Problem	Immediate Action	Escalation Required?
Calibration drift (voltage/ current)	Run daily golden sample check; compare to reference	If drift > spec limit

13. Recommendations

13.1 Logging Improvements

Recommendation	Current State	Improvement
Structured log format (JSON)	Plain text	Enables log aggregation and search with ELK / Grafana
Centralized log shipping	Local files only	Ship to NAS or log server in real-time
Log compression after 7 days	Manual archival	Auto-compress with <code>TimedRotatingFileHandler</code>
Error rate alerting	No alerting	Trigger email/Slack on N errors per hour
DUT serial number in every log line	Inconsistent	Always include <code>serial_number</code> in log context

13.2 Monitoring Improvements

Recommendation	Benefit
Dashboard of daily pass/fail rates per station	Early detection of fixture degradation
RSSI trend chart over time	Detect antenna or RF path degradation before production failures
Supercap charge time trend	Detect PSU capacity reduction
FPA adapter health monitoring (serial number tracking)	Plan FPA replacement proactively

13.3 Production Stability Improvements

Recommendation	Benefit
UPS on all jig PCs	Prevent Excel corruption and mid-test shutdown
USB port locking (Device Manager)	Prevent inadvertent COM number changes
Fixed COM port aliases via registry	Remove dependency on physical USB port order
Pre-production golden sample run (SOP)	Catch configuration drift before shift start
Two-operator sign-off for config changes	Prevent accidental production config edits

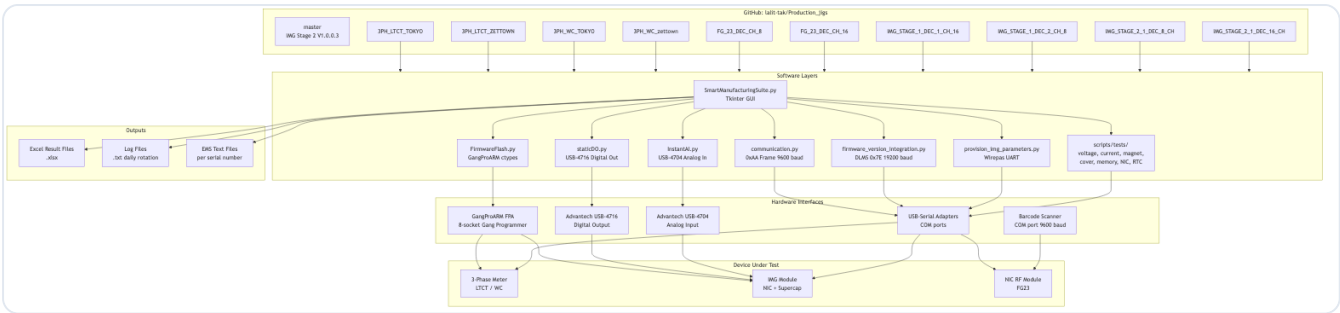
13.4 Error Handling Improvements

Module	Current	Recommended
<code>FirmwareFlash.py</code>	<code>sys.exit(1)</code> on missing firmware	Raise exception, catch in GUI, show user-friendly dialog
<code>communication.py</code>	Returns <code>None</code> on failure	Return typed <code>Result</code> object with error code and message
Excel write	<code>workbook.save()</code> without retry	Implement retry with exponential backoff
Process timeout	Kill + log	Kill + log + mark station with specific <code>TIMEOUT_FAIL</code> result code
<code>serial_config.enc</code> decrypt	Exception propagates	Catch and show "configuration error" dialog with instructions

13.5 Future Automation Opportunities

Opportunity	Description
CI/CD for test scripts	Auto-run unit tests on push to any branch via GitHub Actions
Automated golden sample regression	Schedule nightly golden sample test; compare to baseline
MES integration	POST result JSON to Manufacturing Execution System API
Remote jig health monitoring	PC agent reporting CPU, disk, port health to central dashboard
Automated firmware release management	Version pinning in config JSON with SHA256 checksum verification

Repository Architecture Diagram



Appendix A — GangProARM Error Code Reference

Code	Name	Description
519	<code>STATUS_SELECTED_MCU_METADATA_ERROR</code>	Wrong MCU metadata in config
520	<code>STATUS_SELECTED_WRONG_MCU_ERROR</code>	MCU type mismatch
522	<code>STATUS_JTAG_INIT_ERROR</code>	JTAG probe init failed
523	<code>STATUS_JTAG_COMM_ERROR</code>	JTAG communication failure
525	<code>STATUS_FW_TYPE_ERR</code>	Firmware type mismatch
526	<code>STATUS_FW_VERIFICATION_ERROR</code>	Flash verify failed
535	<code>STATUS_OPEN_FILE_ERROR</code>	Firmware file not found
536	<code>STATUS_FILE_NAME_ERROR</code>	Invalid filename
546	<code>STATUS_FLASH_PRG_VERIFY_ERROR</code>	Flash program verify failure
549	<code>STATUS_ERASE_SEGMENT_FAILED</code>	Flash erase failed
551	<code>STATUS_LOCKED_DEVICE</code>	OTP/security bits locked
554	<code>STATUS_CORRUPT_CONFIG_ERROR</code>	Corrupted <code>.cfg</code> file
990	<code>STATUS_IMAGE_DESTINATION</code>	Image destination error

Full list in `errors_dll.py` (present in FG23 and IMG Stage 2 branches).

Appendix B — Serial Protocol Command Reference (0xAA Frame)

Command	TX Bytes	Expected RX Length	Used In
Get Live Version	<code>AA 01 4E 47 C9 FF FF FF FF 04 04 02 4E 47 E7 0C</code>	17 bytes	<code>live_version_test.py</code>
Get Internal Version	<code>AA 01 4E 47 C9 FF FF FF FF 04 04 02 D3 12 2C 2E</code>	17 bytes	<code>live_version_test.py</code>
Get 3P Voltage	<code>AA 01 4E 47 C9 FF FF FF FF 04 04 02 F2 33 A8 0F</code>	26 bytes	<code>voltage_test.py</code>
Get 3P Current	<code>AA 01 4E 47 C9 FF FF FF FF 04 04 02 F2 34 31 98</code>	29 bytes	<code>current_test.py</code>
Get Magnet Status	<code>AA 01 4E 47 C9 FF FF FF FF 04 04 02 F2 35 02 A9</code>	15 bytes	<code>magnet_status_test.py</code>

Document generated from repository analysis of [lalit-tak/Production_jigs](#) as of 2026-06-15.
For updates, raise a PR against this document or contact the Polaris Grids software team.